

Contest Mindset and Diagnostic

Module 1 · Teacher Lesson Plan · Gauss Contest Preparation

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1. Module header

Field	Value
Module	1 — Contest Mindset, Diagnostic and Toolkit
Hours · week	3.0 h (two 90-minute lessons, no extension) · Week 1
Teaching mode	— (non-content: orientation and diagnosis)
Share of contest points	Non-content module. It places students and installs the tools every later module uses.
Curriculum	Ontario Mathematics (2020), Grade 7 — Strand A: Social-Emotional Learning Skills in Mathematics and the Mathematical Processes (verify expectation codes against the official document). The diagnostic samples Strands B-E.
Contest	Gauss Grade 7: 25 multiple-choice questions, five options (A)–(E), 60 minutes. Part A Q1–10 at 5 points, Part B Q11–20 at 6 points, Part C Q21–25 at 8 points; maximum 150. Confirm calculator and blank-answer rules on the current CEMC contest page before each sitting. Reference: https://cemc.uwaterloo.ca/contests/gauss
Student goal	Know how Gauss is scored, and use six core moves to start any problem.
Files	Entrance Diagnostic Form A + marking key · Persistence Observation Rubric · Student Strategy Card · Error Log Template · Placement Report Template

2. Why this module carries this weight

Three hours, no content points — because every later hour depends on it. The diagnostic places each student on a pathway before content teaching starts, the error log turns every later mistake into data, and the six core moves are the vocabulary the lesson plans for Modules 2–20 use. Teach this for **accurate placement and habits**, not for speed or depth. The one number to land: Parts A and B hold 110 of 150 points (73%).

3. Prerequisites

Earlier modules: none. Grade 6 mathematics is assumed. **Five-minute check** (oral, before the diagnostic): “Have you written a maths contest before? What do you do when you are stuck?” and one warm-up: “What is 25% of 60?” (15). The purpose is to settle nerves and hear how the student talks about being stuck; record the answer on the rubric's evidence notes.

4. Lesson breakdown

Lesson A (90 min) — format, scoring, diagnostic

Min	Phase	What happens
0–10	Welcome and contest format	Section 5a script: what Gauss is, 25 questions, 60 minutes, five options. Student predicts which part is worth most.
10–20	Scoring geometry	Build the 5/6/8 table together; the 73% insight; pacing model from the strategy card.
20–80	Entrance Diagnostic, Form A	60 minutes, calculator-free. You observe and score the persistence rubric (separate sheet). At most one neutral hint per student, recorded.
80–90	Exit reflection	Student writes: easiest question, hardest question, one question they would change their answer on. Collect papers.

Lesson B (90 min) — core moves, error classification, error log

Min	Phase	What happens
0–10	Retrieval	Scoring geometry quiz: What is Part B worth in total? Score for all of A and half of B? Which part holds 73% of the points?
10–25	Concept / strategy	The six core moves (section 5e script), each with its trigger. Hand out the Strategy Card.
25–45	Guided practice	Return the scored diagnostic. Re-work three of the student's missed items together, naming the core move each needed.
45–70	Paired → independent	Error-classification conversation (section 5c) on every remaining wrong item; student fills the first rows of the Error Log. Challenge items 21–25 re-attempted with the hint ladder.
70–80	Method comparison	Diagnostic Q12 or Q19: bar model vs equation, Venn diagram vs list. Which move was faster? Which was safer?
80–90	Exit check	Two items (section 10). Placement conversation outline; parent update sent after class.

Lesson A cannot follow the standard 10/15/20/25/10/10 cycle because the diagnostic needs an unbroken 60 minutes. Lesson B follows the standard cycle.

5. Content sections and scripts

5a. Contest format and the 5/6/8 scoring geometry

Gauss Grade 7: 25 multiple-choice questions, five options (A)–(E), 60 minutes. Part A Q1–10 at 5 points, Part B Q11–20 at 6 points, Part C Q21–25 at 8 points; maximum 150. Confirm calculator and blank-answer rules on the current CEMC contest page before each sitting.

- **Say:** “Gauss has 25 questions and 60 minutes. Before I tell you the points — which part do you think is worth the most?” **Pause.** Most students say Part C.
- **Build the table on the board with the student:** A $10 \times 5 = 50$, B $10 \times 6 = 60$, C $5 \times 8 = 40$. **Ask:** “What fraction of the points is in A and B together?” Expected secure answer: 110 of 150, about three-quarters.

- **Decision point:** if the student still wants to spend most time on Part C, ask: “A perfect Part C and nothing else — what score? A perfect A and half of B?” (40 vs 80).
- **Pacing:** hand out the Strategy Card and read the pacing model aloud: 15 / 25 / 15 / 5 minutes.

5b. Administering and scoring the diagnostic

- Materials: paper, pencils, eraser, scrap paper. No calculator. A clock the student can see.
- **Say exactly:** “This is not a test you pass or fail. It shows me what to teach you first. Do what you can, skip what you can't, and show your thinking in the method boxes — they earn points.”
- Start the timer. Warnings at 30 and 50 minutes only. Observe with the persistence rubric; give at most one neutral hint (rubric sheet protocol).
- Score with the marking key: 2/5/6 points, partial credit only as the key states. Fill the strand profile and total on the placement report before Lesson B.
- Never discuss the score during Lesson A. The first conversation about results is the error-classification conversation in Lesson B.

5c. The error-classification conversation

For each wrong item, follow this order. The student chooses the code; you only ask.

- **Re-attempt cold.** “Try this one again now, no help.” If it is now right, ask “What changed?”: misread → **Q**; calculation → **A**; ran out of time → **T**; “I remembered how” → **K**.
- **Still wrong?** Ask “What is the question asking?” If the student cannot say → **Q**.
- **Can say it but cannot start?** Ask “What could you draw or tabulate?” No idea → **R**; draws it and then solves → **S** (the move was available but not chosen).
- **Has a plan but the idea is wrong** (e.g. adds successive percents) → **C**.
- **Misread or trusted a figure** → **V**.
- Close each item with: “Next time I will...” — the student writes it in the last column of the Error Log.

5d. Setting up the error log

- Give the student the Error Log Template. Model the first row yourself from a diagnostic item.
- Rule: one row per wrong answer, one code per row, and a ‘what I will do differently’ that is an action (“draw a bar model before calculating”), not a feeling (“be more careful”).
- The log is reviewed at the start of every extension session and before every mock (Modules 19–20).

5e. The six core moves — script with original mini-examples

Move	Mini-example (say it, board it)	Model solution	What student answers sound like
Draw it	A 12 m rope is cut into three pieces. The second is twice the first. The third is 2 m longer than the second. How long is the longest piece?	Draw three bars: 1 unit, 2 units, 2 units + 2 m. Five units + 2 = 12, so a unit is 2 m; longest = 6 m.	Not yet: guesses 4 m each. Developing: writes x , $2x$, but stalls on the third. Secure: draws the bars and says “five equal units and an extra 2”.

Move	Mini-example (say it, board it)	Model solution	What student answers sound like
Tabulate it	In how many ways can you make 30¢ using only dimes and nickels?	Table: dimes 0, 1, 2, 3 with nickels 6, 4, 2, 0. Four ways.	Not yet: lists 2–3 at random and stops. Developing: lists in no order and misses one. Secure: orders by dimes and says “I know I have them all because dimes go 0 to 3”.
Try a small case	Ten people each shake hands once with every other person. How many handshakes?	2 people: 1. 3 people: 3. 4 people: 6. Each new person adds one more than the last: $1+2+\dots+9 = 45$.	Not yet: says 100 (10×10). Developing: says 90 (forgets each shake is counted twice). Secure: builds the small cases and explains the +2, +3 pattern.
Work backward	Lia spends half her money, then spends \$4 more, and has \$6 left. How much did she start with?	From \$6: undo ‘spend \$4’ $\rightarrow$ \$10; undo ‘spend half’ $\rightarrow$ \$20.	Not yet: $6 \times 2 + 4 = 16$ (undoes in the wrong order). Developing: gets \$20 but by trial. Secure: reverses each step and checks forward.
Eliminate options	Which could be the product of two consecutive whole numbers? (A) 41 (B) 55 (C) 72 (D) 89 (E) 91	One of two consecutive numbers is even, so the product is even. Only 72 survives (8×9).	Not yet: multiplies pairs at random. Developing: finds 8×9 by search. Secure: eliminates four odd options in five seconds, then confirms.
Check reasonableness	A student writes $19 \times 21 = 299$. Is that reasonable?	19×21 is close to $20 \times 20 = 400$, so 299 is too small. (It is 399.)	Not yet: accepts it. Developing: redoes the long multiplication. Secure: estimates in two seconds and flags it.

For each move: read the problem, **pause 20 seconds**, ask “Which move?”, then let the student run it before you model. Hand the pen over for the last three moves.

6. Misconceptions (probe first, then correct)

Misconception	Probing question	Correction
Part C is where the points are, so spend most time there.	Where are 110 of the 150 points?	Parts A and B hold 110 points (73%). Bank them first; Part C gets a fixed 15 minutes.
If I am not sure, leave it blank.	What do you lose by writing your best elimination guess?	Check the current-year rules on the CEMC page; where a wrong answer costs nothing, never leave a blank — eliminate, then choose.
The diagram shows the answer — I can measure it.	What does ‘Diagrams are not drawn to scale’ mean for Q16?	Use the given numbers and facts, not the look of the picture. Diagnostic figures are to scale, but contest ones may not be.

Misconception	Probing question	Correction
I must do the questions in order.	You are stuck on Q14 at minute 25. What should happen next?	Mark it '?', move on, return in the check window. Order is a suggestion, time is not.
Checking wastes time.	How many points would Q-reading errors have cost you on the diagnostic?	The last 5 minutes are for re-reading marked questions; Q errors are the cheapest points to recover.
A wrong answer means I am bad at maths.	What code would you give this error?	Every error gets a code; codes show what to practise. The log turns mistakes into a plan.
Showing work is only for the teacher.	On Q15, which part of your drawing earned points?	Representations earn method credit here and catch errors in the contest. Draw first, calculate second.

7. Hint ladders for the challenge items (Lesson B review only)

Not used during the diagnostic itself. In Lesson B, give one level at a time; the student records the level used (H1, H2, H3) in the error log.

Item	H1 — reread, name a constraint	H2 — a representation, no calculation	H3 — one intermediate relationship
Diagnostic Q21	Reread: which group changes and which stays the same?	Draw 3 bars for boys and 4 for girls, same size.	The extra 6 boys fill exactly one bar.
Diagnostic Q22	Reread: 'even' — how far apart are the numbers?	Write the five numbers around a middle number m.	Their sum is 5m.
Diagnostic Q23	Reread: exactly two faces — where on a cube does a small cube touch two painted faces?	Sketch one edge of 4 cubes; mark which cubes touch three faces.	Each edge has 2 cubes that are not corners.
Diagnostic Q24	Reread: how many different results can two dice show together?	Build a 6×6 grid of sums.	There are 36 equally likely outcomes; count the 8s.
Diagnostic Q25	Reread: can the first digit be 0?	Make a table with one row for each hundreds digit.	With hundreds digit h, the other two digits add to $5 - h$: that has $6 - h$ ways.

8. Differentiation

Struggling (anxious or below 50 on the diagnostic): in Lesson B re-work only Part 1 misses first, with smaller numbers (e.g. Q12 with 16 marbles), then return to the original. Teach three moves (draw, tabulate, work backward) deeply before the other three.

On-level: full Lesson B; error-classify every wrong item; attempt two challenge items with the hint ladder.

Advanced (85+): every challenge item re-done with a second method and a one-line completeness argument (e.g. why Q25's list has no missing cases); Part C extension: how

many three-digit numbers have digit sum 10?

9. Per-item teacher notes (Entrance Diagnostic Form A)

Full notes, partial-credit rules and representation criteria are in the marking key file. Summary:

Q	Pts	Strand	Answer	Concept	Likely error
1	2	N	(D) 27	Fraction of a quantity	(A) 9 — found $1/4$ only
2	2	N	(B) 12	Percent of a quantity	(E) 65 — subtracted 15 from 80
3	2	A	(C) 9	Solving a one-step-then-two-step equation	(D) 12.5 — added 7 instead of subtracting: $50 \div 4$
4	2	A	(D) 68	Arithmetic sequence: nth term	(E) 75 — used $5 + 10 \times 7$ (one step too many)
5	2	G	(C) 38 cm	Perimeter of a rectangle	(D) 84 cm — found the area
6	2	G	(C) 65°	Angle sum of a triangle	(D) 115° — added the two angles and stopped
7	2	D	(D) 14	Mean of a data set	(B) 12 — gave the median
8	2	T	(A) 59	Primes; divisibility by 3 and 7	(E) 91 — looks prime, but $91 = 7 \times 13$
9	2	T	(E) 24	Lowest common multiple	(A) 2 — gave the greatest common factor
10	2	L	(A) Ben	Logical deduction from constraints	(E) — did not combine the two clues
11	5	N	(B) \$67.80	Successive percent change	(C) \$70.40 — added -25% and $+13\%$ to get -12% of \$80
12	5	N	(B) 12	Ratio with a fixed total (parts of a whole)	(A) 6 — found one part ($48 \div 8$) and stopped
13	5	N	(C) $1\frac{1}{2}$	Fraction of a remainder ('of what')	(B) $5/12$ — took $1/4$ of the whole, not of the rest
14	5	A	(B) 61	Linear pattern from a table	(D) 80 — used 4 per square, ignoring shared sides
15	5	G	(C) 68	Composite area; finding a missing side	(E) 80 — used the full 10×8 rectangle
16	5	G	(B) 42	Angles on a straight line; one-variable equation	(C) 63 — solved $2x + 54 = 180$
17	5	G	(C) 5 cm	Volume of a prism (reverse)	(E) 45 cm — divided by 8 only
18	5	D	(D) 6	Probability as a fraction of the total; reverse	(E) 15 — gave the total, not the green count
19	5	T	(C) 47	Inclusion-exclusion counting	(D) 53 — counted multiples of 15 twice

Q	Pts	Strand	Answer	Concept	Likely error
20	5	L	(D) Dee	Ordering from constraints	(C) Cal — placed Cal before Dee
21	6	N	48	Ratio change when one part changes	42 — the total before the 6 boys joined
22	6	A	38	Consecutive numbers: sum = count $\times$ middle	34 — gave the middle number
23	6	G	24	Spatial counting on a cube's edges	8 — counted the corner cubes (three faces)
24	6	D	5/36	Outcome space for two dice	5/11 — counted sums as equally likely
25	6	T	15	Organised casework; completeness	10 or 21 — missed or double-counted cases (e.g. allowed a leading 0)

10. Exit check and parent update

Lesson A exit: the reflection slip (easiest, hardest, one answer to change). **Lesson B exit:** (1) “Which part of Gauss holds the most points?” (B, 60). (2) “Give the error code: you knew 15% of 80 but wrote 8 because you read it as 10%.” (Q). Both right, and a first error-log row in the student's own words, means the toolkit is installed.

Parent update line: “[Child] completed the Gauss entrance diagnostic and set up an error log; strongest at ____; we will focus first on ____ (placement: ____ pathway).”